B. Rajesh
192526022B

SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**

Course Code: **MLA02**

Course Name: **FUNDAMENTALS OF MACHINE LEARNING**

Duration: **1:30 Hr**

Date : _____

Max Marks: **50 Marks**

ASSIGNMENT – II

UNIT II – SUPERVISED LEARNING

Topics: *Linear regression and classification, naïve Bayes, discriminant functions, logistic regression, decision trees and pruning, neural networks and back-propagation, SVM, ensembles (bagging, boosting)*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO2	Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications.

Answer all the Tasks

Task 1. The following data relate the number of training epochs x to a validation score y.

x	1	2	3	4	5
y	2.1	3.9	6.2	7.8	10.1

(a) Fit $y = w_0 + w_1x$ by ordinary least squares using the normal equations. (b) Predict y at $x = 6$. (c) Compute the residual sum of squares, MSE and R^2. (d) Analyse the reliability of predicting at $x = 6$.

(5) (CO2) (BL4)

Task 2. For mean-centred data with $Sxx = \Sigma x^2 = 10$ and $Sxy = \Sigma xy = 19.9$ (single feature, no intercept needed), compute the ridge-regression slope for $\lambda = 5$. Calculate the percentage shrinkage relative to the least-squares slope and explain the bias–variance effect of increasing λ.

(2) (CO2) (BL4)

Task 3. An e-mail filter uses two binary features: Offer (contains the word 'offer') and Link (contains a hyperlink). Ten labelled e-mails are available.

#	Offer	Link	Class
1	Y	Y	Spam
2	Y	N	Spam
3	Y	Y	Spam
4	N	Y	Spam
5	Y	Y	Spam

#	Offer	Link	Class
6	N	N	Ham
7	N	Y	Ham
8	Y	N	Ham
9	N	N	Ham
10	N	N	Ham

(a) Estimate the priors and the class-conditional probabilities without smoothing. (b) Classify an e-mail with Offer = Y, Link = Y and report the posterior P(Spam | x). (c) Repeat (b) with Laplace smoothing (α = 1). (d) Classify an e-mail with Offer = N, Link = N (with smoothing) and analyse the effect of the conditional-independence assumption and of smoothing.

(6) (CO2) (BL4)

Task 4. A logistic-regression classifier $p = \sigma(w_1x_1 + w_2x_2 + b)$ is initialised with $w = (0, 0)$, $b = 0$ and trained on two samples: $x^{(1)} = (1, 2)$ with $y = 1$, and $x^{(2)} = (2, 1)$ with $y = 0$. Perform one batch gradient-descent step with learning rate $\eta = 0.5$ on the mean cross-entropy loss. Report the gradient, the updated parameters, the predicted probabilities and the loss before and after the update, and analyse the result.

(4) (CO2) (BL4)

Task 5. A node contains 10 training samples (6 Yes, 4 No). A candidate attribute A splits it into $A = a_1$ (4 samples: 4 Yes, 0 No) and $A = a_2$ (6 samples: 2 Yes, 4 No). Calculate the entropy of the node, the information gain of the split and the Gini-impurity decrease, and evaluate whether the split is worthwhile.

(3) (CO2) (BL4)

Task 6. In a decision tree trained on 100 samples, an internal node t with a four-leaf subtree T_t makes 3 training errors, whereas collapsing t into a single leaf would make 8 errors. Using cost-complexity pruning with $R_\alpha(T) = R(T) + \alpha|\text{leaves}|$, compute the critical value α at which the subtree and the single leaf have equal cost, and evaluate whether the subtree should be pruned for $\alpha = 0.01$ and $\alpha = 0.02$.

(3) (CO2) (BL5)

Task 7. A network has two inputs, one sigmoid hidden unit and one sigmoid output unit (no biases): $h = \sigma(w_1x_1 + w_2x_2)$, $\hat{y} = \sigma(vh)$, with $w_1 = 0.4$, $w_2 = 0.3$, $v = 0.5$. For the input $x = (1, 2)$ with target $t = 1$ and squared-error loss $E = \frac{1}{2}(t - \hat{y})^2$, perform one forward pass and one back-propagation update with $\eta = 0.5$. Report h, $\hat{y}$, E, the error terms δ, the gradients, the updated weights and the new loss.

(4) (CO2) (BL4)

Task 8. A linear SVM in two dimensions has $w = (0.5, 0.5)$ and $b = -2$, with the support vectors (3, 3) (class +1) and (1, 1) (class −1). (a) Verify that the support vectors lie on the margin hyperplanes and compute the geometric margin width. (b) Classify the points (4, 2) and (1, 2) and compute their distances from the decision boundary. (c) Compute the hinge loss for (1, 2) with true label −1 and for (2, 2) with true label +1, and state which points violate the margin.

(4) (CO2) (BL4)

Task 9. Independent classifiers, each with accuracy 0.70, are combined by majority voting. Calculate the ensemble accuracy for 3 and 5 classifiers, comment on the gain from adding more classifiers, and analyse what happens to this gain if the classifiers' errors are strongly correlated.

(3) (CO2) (BL4)

Task 10. AdaBoost starts with five training samples of equal weight. The first weak learner misclassifies samples 2 and 5. Compute the weighted error, the learner weight α, the un-normalised and normalised weights after the update, and analyse how boosting changes the focus of the next learner.

(4) (CO2) (BL4)

Task 11. Two classes share the covariance matrix $\Sigma = [[2, 0.5],[0.5, 1]]$ and have means $\mu_1 = (1, 2)^T$ and $\mu_2 = (3, 1)^T$. (a) Derive the discriminant direction $w = \Sigma^{-1}(\mu_2 - \mu_1)$ and the decision threshold for equal priors. (b) Classify $x = (2.5, 1)$. (c) Evaluate whether the decision changes if the priors are $P(\omega_1) = 0.8$ and $P(\omega_2) = 0.2$

(4) (CO2) (BL4)

Task 12. A bank evaluates three classifiers on 1,000 test loans (100 actual defaulters). Default is the positive class.

Model	TP	FN	FP	TN
A	70	30	90	810
B	85	15	180	720
C	60	40	30	870

A missed defaulter (FN) costs ₹50,000 and a wrongly rejected good applicant (FP) costs ₹5,000.

(a) Compute accuracy, precision, recall and F1-score for each model. [3]

(b) Compute the total expected cost of each model. [2]

(c) Critically evaluate which model should be deployed and explain why accuracy and F1 lead to a different choice than cost. [2]

(d) Propose two modelling or threshold-tuning steps to improve the chosen model and one validation safeguard against over-fitting. [1]

(8) (CO2) (BL6)

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ANSWER SCRIPT EVALUATION SHEET

ASSIGNMENT - II

Programme	B.E / B.Tech	**Course Code**	MLA02
Course Name	**Fundamentals of Machine learning.**	**Date**	
Duration	1 Hr 30 Min	**Max Marks**	50

Register No.: _____

TASK 1	Max Marks	Marks Awarded
Task 1	5	4 1/2
Task 2	2	2
Task 3	6	5 1/2
Task 4	4	4
Task 5	3	2 1/2
Task 6	3	3
Task 7	4	3 1/2
Task 8	4	4
Task 9	3	3
Task 10	4	3 1/2
Task 11	4	3 1/2
Task 12	8	7
TOTAL	**50**	46

Total in words:

B. Rajesh
192526026.

1A) Linear regression

(a) fit the regression equation.
using the least squares

$w_1 = 1.99, \quad w_0 = 0.04$

Regression equal

$$y = 0.04 + 1.99x$$

(b) prediction at $x = 6$

$y = 0.04 + 1.99(6) = 11.98$

(c) Performance measure

RSS = 0.028
MSE = 0.0056
$R^2 = 0.9993$

(d) Reliability analysis

* The predictions is reliable because the data follows an almost linear pattern thower $x = 6$ is ootside training range soit is an extrapolation

en) Ridge Regression

Given $S_{xx} = 10$, $S_{xy} = 19.9$, $\lambda = 5$

Ridge slope:

$$w = \frac{19.9}{10+5} = 1.3267$$

least squares slope:

$$w_{ls} = \frac{19.9}{10} = 1.99$$

Percentage shrinkage

$$\frac{1.99 - 1.3267}{1.99} \times 100 = 33.33\%$$

$$= 33.33\ \%$$

sn) Naive Bayes classification

(a) prior Probabilites

$$P(\text{spam}) = 5/10 = 0.5$$

$$P(\text{ham}) = 5/10 = 0.5$$

Conditional Probability

Probability	spam	ham
offer = Y	0.8	0.2
offer = N	0.2	0.5
link = Y	0.8	0.2
link = Y	0.2 .	0.8

(b) offer = y, link = y

$$P(\text{spam}|x) = \frac{0.5(0.8)(0.8)}{0.5(0.8)(0.8) + (0.5)(0.2)(0.2)}$$

$$\boxed{P(\text{spam}|x) = 0.941\dot{2}}$$

class: spam

(c) with laplace smoothing

$$P(\text{offer} = y \mid \text{spam}) = 6/7$$

$$P(\text{link} = y \mid \text{spam}) = 6/7$$

for Ham both Probability are $2/7$

$$P(\text{spam}|x) = 0.90$$

class: spam

(d) offer = N, link = N

$$P(\text{spam}|x) = 0.10$$

class: ham.

*smoothing prevents zero probabilites. The indenpences assumption simplifies claeblations but may ignore relactionship between features

4n) Logistic Regression

Given $\eta = 0.5$ $w_1 = w_2 = 0$, $b = 0$

Intinal Probabilites

$$P_1 = P_2 = 0.5$$

Gradient $\frac{dL}{dw_1} = 0.25$

$\frac{dL}{dw_2} = -0.25$

$\frac{dL}{db} = 0$

updated probabilites :-

- sample 1 = 0.5312
- sample 2 = 0.4688

loss

- Befor update = 0.6931
- after update = 0.6326

analysis :- The loss decreases after updating the weights showing that the model is learning correctly.

Decision tree

Given 10 samples (6 yes, 4 no)

(1) Entropy of parent node

$$H = -0.6\log_2(0.6) - 0.4\log_2(0.4)$$

$$\boxed{H = 0.971}$$

(2) Information gain

Child entropies

$n = a_1 : H = 0$

$n = a_2 : H = 0.9183$

$IG = 0.971 - [0.4(0) + 0.6(0.9183)]$

$$\boxed{IG = 0.420}$$

(3) Gini impurity decrease

Gini(parent) = 0.48

Gini(children) = 0.2669

$$\boxed{\text{Decrease} = 0.2133}$$

Evaluation: The split is worthwhile because it reduces

...ity and improve classification

6 m) Cost Complexity pruning

Given subtree errors = 3, leaves = 4, leaf error

$$3 + 4\alpha = 8 + \alpha$$

$$3\alpha = 5$$

$$\alpha = 1.667$$

Evaluation

- For $\alpha = 0.01$: Do not prune
- For $\alpha = 0.02$: Do not prune

Conclusion: The subtress has lower cost for both value

So it should be retained.

7 m) Neural Network Backpropagation

Given $w_1 = 0.4$, $w_2 = 0.3$, $v = 0.5$, $\eta = 0.5$

(i) Forward propagation

$$h = \sigma(0.4(1) + 0.3(1))$$

$$h = 0.7311$$

$$\hat{y} = \sigma(0.5 \times 0.731(1))$$

$$\hat{y} = 0.5904$$

(ii) error $E = \frac{1}{2}(1 - 0.5904)^2$

$$E = 0.0839$$

3) Error terms

$\delta o = 0.0991$

$\delta h = 0.00974$

(4) Gradients

$v = 0.67242$

$w_1 = 0.00974$

$w_2 = 0.01948$

(5.) updated weights

$v = 0.5362$

$w_1 = 0.4049$ $\quad w_2 = 0.3097$

$\epsilon = 0.0811$

Conclusion: The loss decreases after backpropagation showing improvement in learning

8n) Support vector Machine

Given $w = (0.5, 0.5)$, $b = -2$

(a) margin verification

for (3,3);

$f(x) = 0.5(3) + 0.5(3) - 2 = 1$

for (1,1)

$f(x) = 0.5(1) + 0.5(1) - 2 = -2$

Both lie on margine hyperplanes

$$\text{margin} = \frac{2}{\sqrt{(0.5)^2 + (0.5)^2}}$$

$$\text{margin} = 2.828$$

(b) classification

Point	prediction	Distance
(4,2)	+1	1.414
(2,2)	-1	0.707

(c) Hinge loss

$$L = \max(0, 1 - y\,f(x))$$

- (1,2) true label -1 loss = 0.5
  (2,2) true label +1 loss = 1

conclusion: Both points violate the margin condition

9a) Ensemble learning

given: individual classifier accuracy = 70%

for 3 classifiers.

$$\text{accuracy} = 0.7^3 + 3(0.7^2)(0.3)$$

$$= 78.4\%$$

For 5 classifiers :

$$\text{Accuracy} = \sum_{k=3}^{5} \binom{5}{k} (0.7)^k (0.3)^{5-k}$$

Analysis : ~~nothing~~ classifiers improves accuracy. However strongly correlated errors reduce the benefit of majority voting

) AdaBoost.

Given 5 samples was equal weight

Initial weight

$$w = 1/5 = 0.2$$

Two samples are misclassified

(1) weighted error

$$\epsilon = 2(0.2) = 0.4$$

(2) learner weight

$$\alpha = \tfrac{1}{2} \ln \frac{1-0.4}{0.4}$$

$$\boxed{\alpha = 0.2027}$$

(3) updated weights

- misclassified samples 0.2449 each
- correct samples are 0.16330 each

3/2 Analysis: AdaBoost increases the weights of samples so that the next learner focuses more on difficult sample

11a Discriminant analysis

Given

$$\Sigma = \begin{bmatrix} 2 & 0.1 \\ 0.5 & 1 \end{bmatrix}$$

$$\mu_1 = (1,2) \quad \mu_2 = (3,1)$$

(a) Discriminant direction

$$w = \Sigma^{-1}(\mu_2 - \mu_1)$$

$$w = (1.4286, -1.7143)$$

$$= 0.2857$$

(b) Classification

$x = (2.5, 1)$

$$w^T x = 1.8571$$

Class = w_2

(c) Unequal priors

$$P(w_1) = 0.8, \quad P(w_2) = 0.2$$

1.676

Since 1.8571 is greater than 1.672, the decision remains w_2.

classifier Evalution

Given 1000 test loans and 100 actual defaulters

(a) Performance measure

formulas

$$\text{Accuracy} = \frac{TP + TNT}{\text{Total}}$$

$$\text{Precision} = \frac{TP}{TP + PP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2PR}{P + R}$$

Model	Accuracy	Precision	Recall	F1 score
A	88%	43.75%	70	53.85%
B	80.5%	32.08%	85%	46.58%
C	95%	66.67%	60%	63.16%

Expected Cost

Formula

$$\text{Cost} = (FN \times 50000) + (FP \times 5000)$$

Model	calculation	Total cost
A	80 x 50000 + 90 x 5000	19,50,000
B	15 x 50000 + 180 x 5000	16,50,000
C	46 x 50000 + 30 x 5000	2150,000

(c) Model selection

Model 'B' should be selected because it has the lowest total cost

model 'c' has the highest accuracy & f1 score

(d) Improvement steps

1. Adjust the classifications threshold to reduce missed defaulters
2. Use class weighting to improve default detection

(e) validation safeguard : Use cross validation to Prevent Overfitting